

Stochastic (Random) Process

A process or system which is governed by probability rule is called a stochastic process. Here at least one component is random

Deterministic Process: A process which is governed by someone is known as deterministic Process. Here none of the components are random.

Example of Stochastic Process:

(1) Life of a tube light → Continuous random variable

The process of replacement of bulbs can be a stochastic process.

(2) Queuing Process (3) Birth and Death Process etc.

State: The positions that a stochastic process can occupy are called state.

State space: The set of all possible states/positions that a system can occupy is called state space. State space can be discrete/continuous

Markov Process: [Markov Process is one of the most important class of stochastic process]

The essence of the Markov Property is that, once the state occupied at time n is given, the state occupied before n has no effect on the future behavior.

In Mathematically,

$$\begin{aligned} P(X_{n+1} \mid X_0, X_1, \dots, X_{n-1}, X_n) \\ = P(X_{n+1} \mid X_n) \end{aligned}$$

The process $X(t)$, $t \geq 0$ or X_n , $n=0,1,2,\dots$ Is a stochastic Process. $X(t)$ or X_n may be independent or dependent.

[X_t = When time is discrete; $X(t)$ = When time is Continuous; X_n = most recent known value of the process]

In probability theory, a Markov chain is a stochastic process that refers to the sequence (or chain) of states such a process moves through (Baier, *et al*, 1999). The changes of state of the system are called transitions, and the probabilities associated with various state-changes are called transition probabilities. The process is characterized by a state space, a transition matrix describing the probabilities of particular transitions and an initial state or initial distribution across the state space.

Markov chain(MC): Let us consider parameter and state space discrete. Markov processes with discrete time and state spaces are called Markov Chain.

A markov chain is a sequence of discrete random variables $(X_0, X_1, \dots, X_n)$ with the property that the conditional probability of X_{n+1} , given $X_0, X_1, \dots, X_n$ depends only on the value X_n but not further on $X_0, X_1, \dots, X_{n-1}$.

Mathematically, $X_n, n \geq 0$ is a Markov chain if

$$\begin{aligned} P[X_{n+1}=j \mid X_0=x_0, X_1=x_1, \dots, X_n=i] & \quad ; \quad i,j=0,1,2,\dots \\ &= P(X_{n+1}=j \mid X_n=i) \\ &= P_{ij} \quad [P_{ij} \rightarrow \text{the system is at } j \text{ given that it was at } i \text{ one step before (this is a transition)}] \end{aligned}$$

Or, we can write

$$\begin{aligned} P[X_n=j \mid X_0=x_0, X_1=x_1, \dots, X_{n-1}=i] & \quad ; \quad i,j=0,1,2,\dots \\ &= P(X_n=j \mid X_{n-1}=i) \\ &= P_{ij} \end{aligned}$$

Here P_{ij} is called the one-step transition probability. Transition Probability is nothing but conditional probability.

##n-step transition Probability: Let $X_n, n \geq 0$ be a Markov chain .Then n-step TP is given by

$$P^{(n)}_{ij} = P(X_n=j \mid X_0=i)$$

Here n-step refers to the time interval between observations.

Transition matrix(TM): The matrix with elements equal to transition probability is called TM of MC. Thus

One step transition matrix, $P = (P_{ij}) =$

$$\begin{bmatrix} P_{00} & P_{01} & P_{02} & \dots\dots\dots \\ P_{10} & P_{11} & P_{12} & \dots\dots\dots \\ \cdot & & & \\ \cdot & & & \\ \cdot & & & \end{bmatrix}$$

Order of the transition matrix may be finite or infinite depending on the state space.

Transition matrix is also called stochastic MATRIX

Property of TM:

- (1) TM is a square matrix.
- (2) Since elements of the TM are transition probabilities (conditional probability), the elements are non-negative and $0 \leq P_{ij} \leq 1$.
- (3) All the row totals must be one. If all column totals are also one (which is not compulsory) the TM is called doubly stochastic.

Role of TM: TM places the central role in the analysis of MC. If TM is known, everything (i.e. property) about the MC is known.

EX:

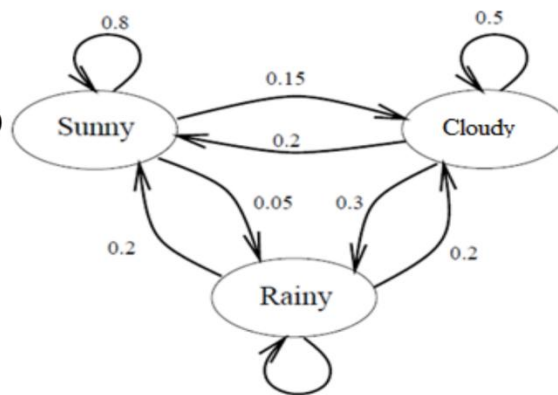
	Dry	Wet	Total
Dry	1049	350	1399
Wet	351	687	1038
Total	1400	1037	2437

- (a) Find the transition Matrix. (b) Find the probability that December 10 is a wet day given that December 7 was a dry day.

- Exercise 1: Given that today is Sunny, what's the probability that tomorrow is Sunny and the next day Rainy?

$$P(q_2, q_3 | q_1) = P(q_2 | q_1) P(q_3 | q_1, q_2)$$

$$\begin{aligned} &= P(q_2 | q_1) P(q_3 | q_2) \\ &= P(\text{Sunny} | \text{Sunny}) P(\text{Rainy} | \text{Sunny}) \\ &= (0.8)(0.05) \\ &= 0.04 \end{aligned}$$



- A communication satellite is launched via a booster system that has a discrete-time guidance control system. Course correction signals form a sequence $\{X_n\}$ where the state space for X is as follows:
 0: No correction required
 1: Minor correction required
 2: major correction required
 If $\{X_n\}$ can be modeled as a Markov chain with one-step transition matrix as

$$\begin{pmatrix} 1 & 0 & 0 \\ \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 1 \end{pmatrix}$$

If today has no required for correction in communication satellite launched, find the probability that it will be required to do major correction exactly 2 days hence.

Q 2: Weather data are analyzed for a particular locality, and a Markov chain is employed as a model for weather change as follows. The conditional probability of change from rain to clear weather in one day is 0.3 likewise. The conditional probability of transition from clear to rain in one day is 0.1. The model is to be a discrete time model, with transition occurring only between days.

- Determine the matrix P of one step transition probability.

(b) Find the probability that the first passage from a clear day to a rainy day occurs in exactly 2 days, given a clear day is the initial state. (Ans: 0.16)

(c) If today is clear, find the probability that it will be clear exactly 3 days hence. (Ans: 0.80)

3. Question: Here is a matrix P of one step transition probability. Now find the probability

of the cure in next 3 days, if today is sick. $P = \begin{pmatrix} 0.3 & 0.1 & 0.6 \\ 0.2 & 0.7 & 0.1 \\ 0.5 & 0.3 & 0.2 \end{pmatrix}$

Let, 0=sick ; 1= Under treatment; 2= Cure

Ans: 0.34

Two-step Transition Probabilities for the Weather Example

- Intuition: to go from state 0 to 0 in two steps we can either
 - Go from 0 to 0 in one step and then go from 0 to 0 in one step OR
 - Go from 0 to 1 in one step and then go from 1 to 0 in one step
- Therefore, $p_{00}^{(2)} = P(X_2 = 0 \mid X_0 = 0) = p_{00}p_{00} + p_{01}p_{10}$
- In short,

$$p_{00}^{(2)} = \sum_{k=0}^1 p_{0k}p_{k0}$$

$$P^{(2)} = \begin{bmatrix} p_{00}^{(2)} & p_{01}^{(2)} \\ p_{10}^{(2)} & p_{11}^{(2)} \end{bmatrix} = \begin{bmatrix} p_{00}p_{00} + p_{01}p_{10} & p_{00}p_{01} + p_{01}p_{11} \\ p_{10}p_{00} + p_{11}p_{10} & p_{10}p_{01} + p_{11}p_{11} \end{bmatrix}$$

- Interpretation : $p_{01}^{(2)}$ is the probability that the weather day after tomorrow will be rainy if the weather today is sunny.
- An interesting observation: the two-step transition matrix is the square of the one-step transition matrix!!! That is,

$$P^{(2)} = P^2$$

- Why?

Recall matrix product to write down P^2 and confirm that it equals $P^{(2)}$ above.

- For a general Markov chain with states $0, 1, \dots, M$, to make a two-step transition from i to j , we go to **some** state k in one step from i and then go from k to j in one step. Therefore, the two-step transition probability matrix is, $P^{(2)} = P^2$

$$P^{(2)} = \begin{bmatrix} p_{00}^{(2)} & p_{01}^{(2)} & \cdots & p_{0M}^{(2)} \\ p_{10}^{(2)} & p_{11}^{(2)} & \cdots & p_{1M}^{(2)} \\ \vdots & \vdots & & \\ p_{M0}^{(2)} & p_{M1}^{(2)} & \cdots & p_{MM}^{(2)} \end{bmatrix} \quad \text{with } p_{ij}^{(2)} = \sum_{k=0}^M p_{ik} p_{kj}$$

n-step Transition Probabilities for General Markov Chains

- For a general Markov chain with states $0, 1, \dots, M$, the **n-step** transition from i to j means the process goes from i to j in n time steps
- Let m be a **non-negative integer not bigger than n** . The Chapman-Kolmogorov equation is:

$$p_{ij}^{(n)} = \sum_{k=0}^M p_{ik}^{(m)} p_{kj}^{(n-m)}$$

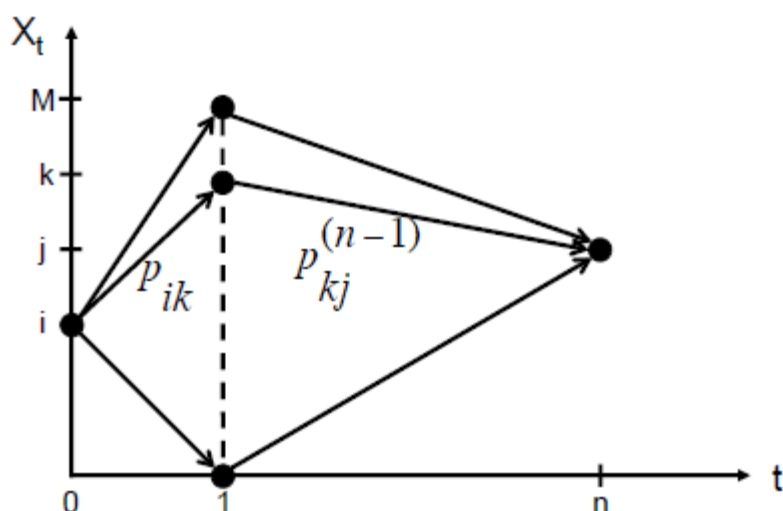
- Interpretation: if the process goes from state i to state j in n steps then it must go from state i to **some** state k in m (less than n) steps and then go from k to j in the remaining $n-m$ steps.
- In matrix notation, $P^{(n)} = P^{(m)} P^{(n-m)}$. This implies that the n -step transition matrix is the n^{th} power of the one-step transition matrix (Why? - substitute $m=1$ and see what happens!)

Chapman-Kolmogorov Equations

$$p_{ij}^{(n)} = \sum_{k=0}^M p_{ik}^{(m)} p_{kj}^{(n-m)} \quad \text{for all } i, j, n \text{ and } 0 \leq m \leq n$$

- Consider the case when $m = 1$:

$$p_{ij}^{(n)} = \sum_{k=0}^M p_{ik} p_{kj}^{(n-1)} = P \cdot P^{(n-1)}$$



- Note, though, that

$$\begin{aligned}
 P^{(n)} &= P \cdot P^{(n-1)} \\
 &= P \cdot P \cdot P^{(n-2)} \\
 &\quad \vdots \\
 &= P \cdot P \cdot P \cdots P = P^n
 \end{aligned}$$

(Unconditional) Probability in state j at time n

- The transition probability $p_{ij}^{(n)}$ is a conditional probability, $P(X_n=j \mid X_0=i)$
- How do we “un-condition” the probabilities?
- That is, how do we find the (unconditional) probability of being in state j at time n, $P(X_n=j)$?
- The probabilities $P(X_0=i)$ define the initial state distribution

$$\begin{aligned}
 P(X_n = j) &= \sum_{i=0}^M P(X_n = j | X_0 = i) P(X_0 = i) \\
 &= \sum_{i=0}^M p_{ij}^{(n)} P(X_0 = i)
 \end{aligned}$$

MARKOV CHAINS



THINK ABOUT IT

If we know the probability that the child of a lower-class parent becomes middle-class or upper-class, and we know similar information for the child of a middle-class or upper-class parent, what is the probability that the grandchild or great-grandchild of a lower-class parent is middle- or upper-class?

Using Markov chains, we will learn the answers to such questions.

A *stochastic process* is a mathematical model that evolves over time in a probabilistic manner. In this section we study a special kind of stochastic process, called a *Markov chain*, where the outcome of an experiment depends only on the outcome of the previous experiment. In other words, the next **state** of the system depends only on the present state, not on preceding states. Applications of Markov chains in medicine are quite common and have become a standard tool of medical decision making. Markov chains are named after the Russian mathematician A. A. Markov (1856–1922), who started the theory of stochastic processes.

Table 1

		Next Generation		
		1	2	3
Current Generation	1	0.65	0.28	0.07
	2	0.15	0.67	0.18
	3	0.12	0.36	0.52

The symbol p_{ij} will be used for the probability of transition from state i to state j in one generation. For example, p_{23} represents the probability that a person in state 2 will have offspring in state 3; from the table above,

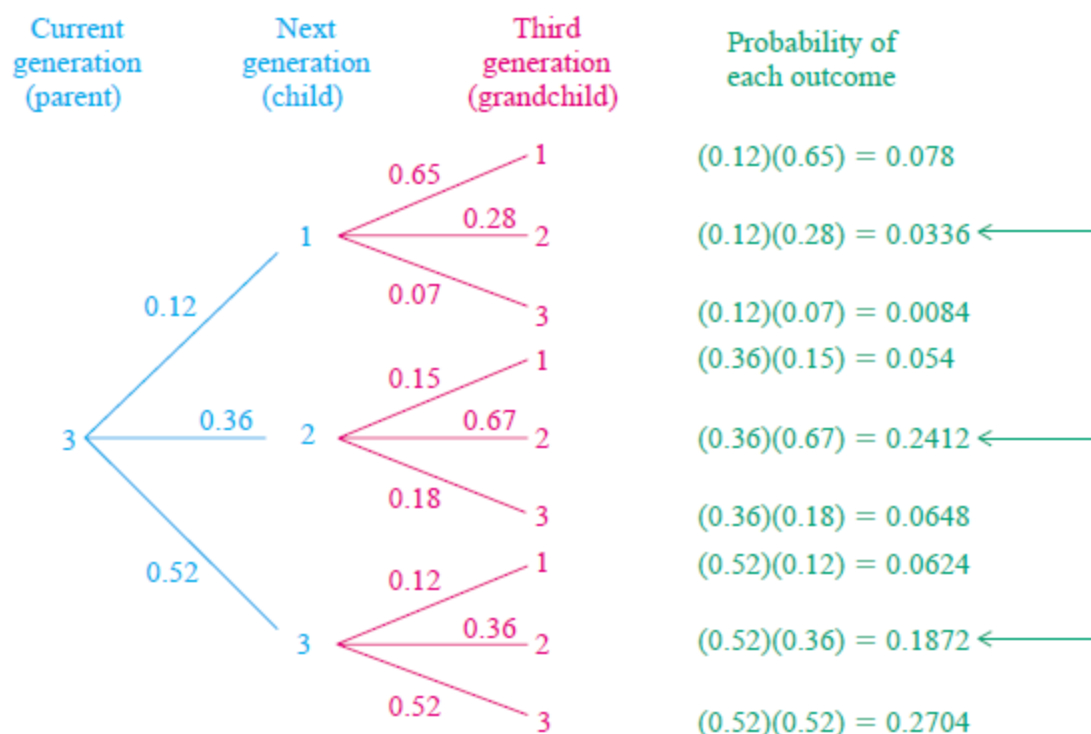
$$p_{23} = 0.18.$$

In a **transition matrix**, the states are indicated at the side and the top. If P represents the transition matrix for the table above, then

$$\begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0.65 & 0.28 & 0.07 \\ 0.15 & 0.67 & 0.18 \\ 0.12 & 0.36 & 0.52 \end{bmatrix} = P.$$

The transition matrix P shows the probability of change in income class from one generation to the next. Now let us investigate the probabilities for changes in income class over *two* generations. For example, if a parent is in state 3 (the upper-income class), what is the probability that a grandchild will be in state 2?

$$0.0336 + 0.2412 + 0.1872 = 0.4620.$$



We used p_{ij} to represent the probability of changing from state i to state j in one generation. This notation can be used to write the probability that a parent in state 3 will have a grandchild in state 2:

$$p_{31} \cdot p_{12} + p_{32} \cdot p_{22} + p_{33} \cdot p_{32}.$$

P^k gives the probabilities of a transition from one state to another in k repetitions of an experiment.

EXAMPLE 1 Transition Matrices

For transition matrix P (income-class changes),

$$P^2 = \begin{bmatrix} 0.65 & 0.28 & 0.07 \\ 0.15 & 0.67 & 0.18 \\ 0.12 & 0.36 & 0.52 \end{bmatrix} \begin{bmatrix} 0.65 & 0.28 & 0.07 \\ 0.15 & 0.67 & 0.18 \\ 0.12 & 0.36 & 0.52 \end{bmatrix} \approx \begin{bmatrix} 0.47 & 0.39 & 0.13 \\ 0.22 & 0.56 & 0.22 \\ 0.19 & \mathbf{0.46} & 0.34 \end{bmatrix}.$$

Row 1, column 3 of P^2 gives the number 0.13, the probability that a person in state 1 will have a grandchild in state 3; that is, that a lower-class person will have an upper-class grandchild. How would the entry 0.47 be interpreted?

EXAMPLE 2 Powers of Transition Matrices

In the same way that matrix P^2 gives the probability of income-class changes after *two* generations, the matrix $P^3 = P \cdot P^2$ gives the probabilities of change after *three* generations.

For matrix P ,

$$P^3 = P \cdot P^2 = \begin{bmatrix} 0.65 & 0.28 & 0.07 \\ 0.15 & 0.67 & 0.18 \\ 0.12 & 0.36 & 0.52 \end{bmatrix} \begin{bmatrix} 0.47 & 0.39 & 0.13 \\ 0.22 & 0.56 & 0.22 \\ 0.19 & 0.46 & 0.34 \end{bmatrix} \approx \begin{bmatrix} 0.38 & 0.44 & 0.17 \\ 0.25 & 0.52 & 0.23 \\ 0.23 & 0.49 & 0.27 \end{bmatrix}.$$

(The rows of P^3 don't necessarily total 1 exactly because of rounding errors.) Matrix P^3 gives a probability of 0.25 that a person in state 2 will have a great-grandchild in state 1. The probability is 0.52 that a person in state 2 will have a great-grandchild in state 2.